

The empirical recurrence for OEIS A302744, recovered from its tail

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8 September 2026

Abstract

OEIS A302744 records “Empirical recurrence of order 64” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 122$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 66$. Its last coefficient is $c_{64} = 992318976 \neq 0$, so no further tail satisfies anything shorter; any order-64 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A302744 is “Number of $5 \times n$ 0..1 arrays with every element equal to 0, 1, 2, 3 or 4 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

16, 512, 11980, 286631, 6947036, 168799572, 4104115772, 99818649293, ...

The entry states:

Empirical recurrence of order 64 (see link above)

As of the “Last modified” line on the live entry (Apr 12 2018 11:24:19, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 122$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 122$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 64: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 64.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 244$ exact terms from $n = 2$ on, it returns order 64 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{64} c_i a(n-i),$$

c_1	24	c_{17}	−27723266168	c_{33}	−189861149201628	c_{49}	51004583328903
c_2	75	c_{18}	−60047532125	c_{34}	−153462157272129	c_{50}	35583305034382
c_3	−1596	c_{19}	107563623499	c_{35}	127623723079590	c_{51}	−44174098406264
c_4	−1829	c_{20}	−185356236023	c_{36}	5948188160107	c_{52}	−2304363301936
c_5	23436	c_{21}	894109335820	c_{37}	100352106949233	c_{53}	−13340173296912
c_6	−29209	c_{22}	1853059238082	c_{38}	74200521373954	c_{54}	−849621280624
c_7	208929	c_{23}	−3587044480716	c_{39}	−151223493162486	c_{55}	1018526970936
c_8	740795	c_{24}	−1771510409104	c_{40}	2848331996830	c_{56}	125359228056
c_9	−6986168	c_{25}	−8955846341817	c_{41}	−870949182149697	c_{57}	−1041784980304
c_{10}	3384599	c_{26}	−13256639073719	c_{42}	496673687197609	c_{58}	−413730854032
c_{11}	20080920	c_{27}	41381946846484	c_{43}	−87324273773381	c_{59}	−311986330576
c_{12}	−123954890	c_{28}	8500408652733	c_{44}	617275044599908	c_{60}	−80334690176
c_{13}	508280233	c_{29}	38761001849443	c_{45}	623761261116420	c_{61}	−35413400704
c_{14}	−25883791	c_{30}	69262640874344	c_{46}	119148888139380	c_{62}	−3774690496
c_{15}	−1663298076	c_{31}	−162649192442047	c_{47}	361954532300696	c_{63}	2987595904
c_{16}	11248873410	c_{32}	51293740763381	c_{48}	37861775735849	c_{64}	992318976

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 66$, and it is the recurrence OEIS A302744 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{64} - c_1 x^{64-1} - \dots - c_{64}$. Its constant term is $-c_{64} = -992318976$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component

of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 64 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 64, so $\deg q = 64$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 64, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 15 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 64, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 992318976.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-64 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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